

Agent360: A Concealment-First Bilateral Negotiator for ANL 2026

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The Goal

ANL SCORE = ADVANTAGE + CONCEALING

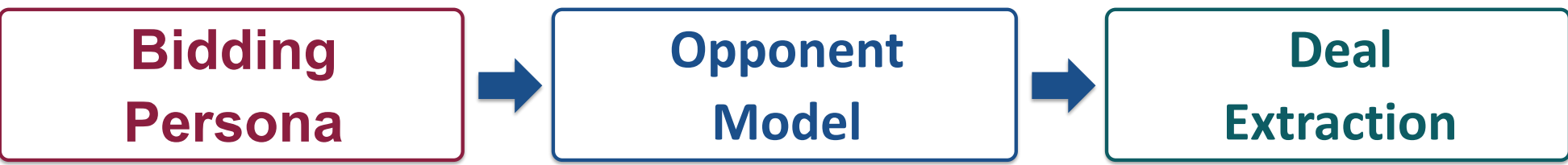
Close good deals above reserved value

Hide true preferences from the opponent's learner

Design principle: concealment first, then extract the deal

Architecture

Bid stream only - no opponent class, no true utility



Bidding Persona



Seat asymmetry: opening seat keeps a longer decoy and stricter gates

Acceptance

Catastrophe floor near deadline

Mode aspiration - slope follows opponent mode

AC-next - as good as our next concealing bid

Bait guard - block deceptive utility spikes

What We Rejected

- Reverse-psychology (truth early) hurts Concealing vs learners
- Mid-game escape accept took mediocre / deceptive deals

Why

Concealment must be maintained until we have strong deal.

Conceal first, then extract.

Opponent Model

Inference uses only the bid stream, relative time, and seat.

Never read opponent class or true utility.

Base: Smith frequency model - value counts per issue → normalized utility estimate.

Recency-blended Smith

On the second seat, blend full-history Smith with a short recent window so late persona shifts are not ignored.

Time-weighted Smith

Late-phase bids count more heavily, reducing contamination from early decoy posturing on either side.

Issue-weighted Smith

Issues with repeated values are down-weighted; diverse issues get higher weight.

Offer trajectory

Track concession slope, non-monotonic rises and predict utility.

Mode Response

- Conceding - Early decoy exit, chase agreement
- Learner - Full blend, no bait discount
- Deceptive - Discount Smith, reject bait
- Mirro - Plain Smith, anti-mirror bids
- Unknown - Conservative full blend

Key Idea

Match the opponent mode and adjust concealment strength accordingly.

Evaluation

	Learners	Stress	Sparring
	BOA · MAP · MiCRO	Time-based NegMAS baselines	mirror · bait-switch · strong learners
wins	48%	95%	63%

Overall 71% wins in in-house evaluations, in the ANL 2026 qualification the agent reached the 32nd place and didn't reach the finals.

Result pattern: strong Concealing against stress agents; competitive balance on learner and deceptive panels. Promoted: early decoy exit, mode closing caps, first-seat decoy rotation, anti-mirror pools, catastrophe accept. Removed after regressions: reverse-truth persona, mid-game escape acceptance, overly strict first-seat gates.

Conclusions

Concealment first

Decoy early and reveal later.

Seat asymmetry

Improving score against learners match ups

Test across panels

Robust on learners, stress and deceptive agents

Separate controls

For concealing and acceptance for bigger advantage.